



Statistical Techniques

Unit No 4: Regression Analysis

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Syllabus

Unit IV: Regression Analysis

-Concept of Regression

-Importance of Regression Analysis in Computer Application.

-Regression Equations:

- Regression equation of Y on X and Regression Equation of X on Y,
- Regression Coefficient



Unit No 4 : Regression Analysis

Concept of Regression



Meaning of Regression

Regression is a statistical technique used to study the relationship between variables and predict one variable based on another.

What does Regression do?

It helps answer questions like:

How does **X** affect **Y**?

Can we **predict Y** if we know **X**?

Example:

If advertising increases, will sales increase?

Regression helps find that relationship.



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Key Idea

- One variable is **dependent (Y)** → what we want to predict
- One or more variables are **independent (X)** → factors influencing Y



Why is Regression Important?

- Helps in prediction & forecasting
- Identifies relationships between variables
- Supports data-driven decision making



Importance of Regression Analysis in Computer Applications



Importance of Regression Analysis in Computer Applications

Regression analysis plays a key role in computer-related fields by helping systems **analyze data, make predictions, and improve decision-making.**

1. Data Prediction

Used to predict future outcomes based on past data

Example: Predicting website traffic or user behavior

2. Machine Learning Models

Core technique in **machine learning algorithms**

Helps systems learn patterns and relationships in data

3. Performance Optimization

Improves system efficiency

Example: Predicting server load to manage resources better



Importance of Regression Analysis in Computer Applications

4. Data Analysis & Visualization: Helps identify trends and relationships

Used in dashboards and analytics tools

5. Business Intelligence: Supports decision-making in software systems

Example: Sales forecasting, customer analysis

6. Risk Management: Helps detect risks and anomalies

Example: Fraud detection systems

7. Software Testing & Quality Assurance: Predicts defects and system failures

Improves software reliability

8. Artificial Intelligence Applications: Used in AI systems for predictions

Example: Recommendation systems (movies, products)

Conclusion

Regression analysis is essential in computer applications because it enables **prediction, optimization, and intelligent decision-making**, making systems smarter and more efficient.



Regression Equations (Y on X and X on Y)

In regression, we often form **two equations**:

Regression of Y on X → used to predict **Y from X**

Regression of X on Y → used to predict **X from Y**

◆ 1. Regression Equation of Y on X

Formula:

$$Y - \bar{Y} = b_{yx}(X - \bar{X})$$

or

$$Y = a + bX$$

Where:

- $b_{yx} = r \cdot \frac{\sigma_Y}{\sigma_X}$



◆ 2. Regression Equation of X on Y

Formula:

$$X - \bar{X} = b_{xy}(Y - \bar{Y})$$

or

$$X = a + bY$$

Where:

- $b_{xy} = r \cdot \frac{\sigma_X}{\sigma_Y}$



Example 1

Given data:

Mean of $X = 10$

Mean of $Y = 20$

Standard deviation of $X = 2$

Standard deviation of $Y = 4$

Correlation (r) = 0.5

find Régression Coefficient (Y on X) and Regression Coefficient (X on Y)



Solution:

✓ Step 1: Regression Coefficient (Y on X)

$$b_{yx} = r \cdot \frac{\sigma_Y}{\sigma_X} = 0.5 \cdot \frac{4}{2} = 1$$

Equation:

$$Y - 20 = 1(X - 10)$$

$$Y = X + 10$$

✓ Step 2: Regression Coefficient (X on Y)

$$b_{xy} = r \cdot \frac{\sigma_X}{\sigma_Y} = 0.5 \cdot \frac{2}{4} = 0.25$$

Equation:

$$X - 10 = 0.25(Y - 20)$$

$$X = 0.25Y + 5$$



Example 2

Given:

Mean of $X = 5$, Mean of $Y = 10$

$\sigma_x = 2$, $\sigma_y = 4$

$r = 0.6$

find Régression Coefficient (Y on X) and Regression Coefficient (X on Y)



Solution:

◆ Step 1: Y on X

$$b_{yx} = r \cdot \frac{\sigma_Y}{\sigma_X} = 0.6 \cdot \frac{4}{2} = 1.2$$

$$Y - 10 = 1.2(X - 5)$$

$$Y = 1.2X + 4$$

◆ Step 2: X on Y

$$b_{xy} = 0.6 \cdot \frac{2}{4} = 0.3$$

$$X - 5 = 0.3(Y - 10)$$

$$X = 0.3Y + 2$$



Example 3

A machine learning researcher studies **GPU utilization (%) (X)** and **model training time (minutes) (Y)**.

GPU Usage (X)	Training Time (Y)
25	18
30	22
35	26
40	30
45	34

Find the regression equation of Y on X and X on Y



Example 4

A data analytics company studies the relationship between **number of machine learning models trained (X)** and **prediction accuracy score (Y)** achieved by interns during training.

Models Trained (X)	Accuracy Score (Y)
5	60
10	65
15	70
20	75
25	80



Questions

1. Find the regression equations of

a) Y on X

b) X on Y

2. Prediction Question

If an intern trains 18 machine learning models, estimate the accuracy score they might achieve.

3. Decision-Making Question

The company wants interns to achieve at least 72 accuracy score.

Using the regression model, how many models should an intern train to reach this target?



Example 5

A software company studies the relationship between **hours spent on coding practice (X)** and **number of bugs fixed in a software project (Y)** by MCA interns.

Coding Practice Hours (X)	Bugs Fixed (Y)
2	5
4	7
6	9
8	11
10	13



Questions

1. Find the regression equations of

a) Y on X

b) X on Y

2. Prediction Question

If an intern practices 7 hours of coding, estimate how many bugs they can fix.

3. Decision-Making Question

The project manager wants each intern to fix 10 bugs per day.

Using the regression model, how many hours of coding practice should be recommended to interns?



Example 6

A company studies **internet usage (minutes) (X)** and **tasks completed (Y)**.

Internet Usage (X)	Tasks Compl (Y)
15	8
20	10
25	12
30	14
35	16



Questions

1. Find the regression equation of tasks completed (Y) on internet usage (X) and internet usage (X) on tasks completed (Y).
2. Predict the number of tasks completed if internet usage is 28 minutes.
3. If a manager wants employees to complete 15 tasks, how many minutes of internet usage should be allowed according to the regression model?



Thank You



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